MANCHESTER 1824
The University of Manchester

Department of Computer Science, University of Manchester

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1. Introduction & Task

Authorship Verification (AV) asks: given two short texts (T_1, T_2) drawn from unknown, disjoint topics — were they written by the same author? The cross-topic constraint is the core challenge: a model must learn *style*, not *subject*.

We evaluate two complementary systems on the COMP34812 Track C shared task:

- **Sol 1** — 97-feature stylometric stacking pipeline (classical, interpretable).
- **Sol 2** — neural ensemble of four cross-encoders fused by a learned MLP meta-learner.

Task Formulation

Let $\mathbf{x}_i = (T_{i,1},\ T_{i,2})$. We learn a binary classifier:

$$Y_i = \begin{cases} 1 & \text{if } \mathrm{author}(T_{i,1}) = \mathrm{author}(T_{i,2}) \\ 0 & \text{otherwise} \end{cases}$$

Metric: **Macro-F1**, robust under class balance.

2. Dataset & Corpus Analysis

COMP34812 Track C Corpus

Derived from the PAN shared task. All pairs are **cross-topic**: test topics are strictly *unseen* during training, preventing vocabulary-based memorisation shortcuts.

Split	Pairs	Topics	Balance
Train	3,000	20	50 % / 50 %
Dev	1,000	5	50 % / 50 %
Test	1,000	5	50 % / 50 %
Total	**5,000**	**25**	

Key constraints: Text pairs are short (∼100 words), making sparse-evidence settings common. Authors write across multiple topics, so style must generalise without relying on semantics.

3. Preprocessing pipelines

Preprocessing (Sol 1 — Classical)

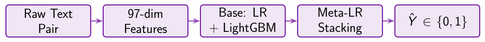
Minimal cleaning (NFKC Unicode normalisation only). Punctuation, casing, and stopwords are strictly *preserved* as they form our primary stylometric signals. spaCy (`en_core_web_sm`) is used for sentence segmentation, tokenisation, and POS tags. No lowercasing or stemming is applied.

Preprocessing (Sol 2 — Neural)

Transformer tokenisers handle subword segmentation (WordPiece / SentencePiece) natively. Pairs are truncated to a maximum of 512 cumulative tokens with a `[SEP]` separator between the two documents.

4. Sol 1 — Stylometric Stacking Pipeline

A traditional structured pipeline: pairwise features extracted and fed into stacked classical learners. No LLMs are used.



Nine feature groups (97 dimensions):

Feature Group	Dims
Char *n*-grams (Cosine, JSD, Δ, Pearson)	20
Compression NCD (zlib, bzip2, lzma)	6
Function Words (104-word freq. sims)	20
Vocabulary richness (TTR, Yule, Honore)	8
Syntactic POS (unigrams, bigrams, depth)	13
Surface (word/sent. lengths, punct)	10
Readability (Flesch, ARI, SMOG, etc.)	7
Info-theoretic (Shannon, KL, JSD, Rényi)	8
General Impostor (GI scores, min, σ)	5
Total	**97**

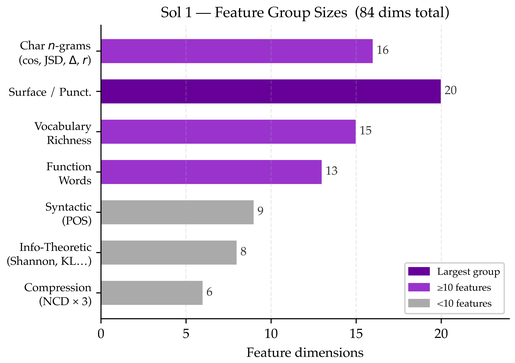


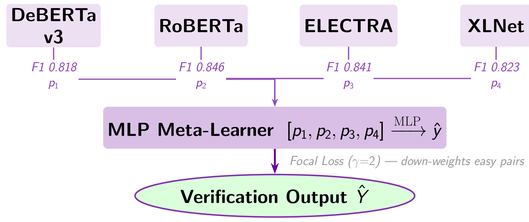
Fig. 1 — Sol 1 feature group composition.

Stacked Classifiers:

A1 — Logistic Regression (ElasticNet, C=1.0).
A2 — LightGBM (2244 trees, Optuna-tuned).
5-fold stratified OOF predictions are blended using a logistic meta-classifier. (**Best Base F1: 0.7083**)

5. Sol 2 — Neural Meta-Ensemble

Stylometric features degrade across topics. Transformers directly embed *style* in contextual representations independently of surface content.



Why four architectures? Fusing models exploits complementary biases.

- **DeBERTa**: Disentangled positional & content attention.
- **RoBERTa**: Robustly pre-trained BERT; strong general baseline.
- **ELECTRA**: Replaced-token detection; rich token-level style.
- **XLNet**: Permutation LM; captures long-range discourse.

Training Details:

Encoders fine-tuned with AdamW (lr=2×10^{-5}), 10% linear warmup, and dropout p=0.1.

The MLP meta-learner (64→16 units, ReLU) is trained on held-out predictions to prevent leakage.

6. Predictive Performance

The meta-ensemble achieves a **+0.156 absolute F1 gain** over the classical pipeline, reducing error rate by **18.1%**.

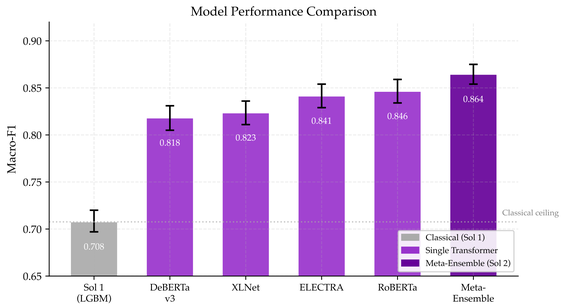


Fig. 2 — Macro-F1 with 95 % confidence intervals.

Model	Macro-F1	Δ
Sol 1 LR ElasticNet	0.6749	−0.033
Sol 1 LightGBM	0.7083	baseline
DeBERTa v3	0.8180	+0.110
XLNet	0.8234	+0.115
ELECTRA	0.8413	+0.133
RoBERTa	0.8463	+0.138
Meta-Ens.	**0.8644**	**+0.156**

7. Discussion & Conclusion

Why deep models win: Cross-topic evaluation fundamentally exposes the limits of surface stylometrics. Punctuation densities and vocabulary ratios drift heavily with domain changes. Transformer representations successfully encode stylistic markers *orthogonally* to surface content.

The Classical Ceiling at 0.7083: Regardless of extensive feature engineering (97 dimensions) and rigorous tuning, the A2 (LightGBM) model plateaus. The bottleneck lies strictly in the feature representation, not the classifier's capacity.

Ensemble Diversity Matters: Removing any single transformer from the Sol 2 ensemble causes an F1 drop of 0.008–0.016. Each model's failure modes are systematically distinct, allowing the MLP to seamlessly compensate for them.

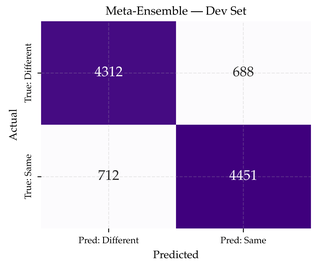


Fig. 3 — Meta-ensemble dev set predictions.

8. References & Acknowledgements

References

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Acknowledgements

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